

流式工单下发接口说明

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一、接口说明

本接口用于将 Vita 中已编辑、并完成业务校验的流式实验工单，以 JSON 形式同步下发给镁伽系统。

镁伽系统接收后，据此安排样本板、细胞板及对照信息的上机执行。

说明范围仅限「工单下发」这一次请求/响应。运行状态回传、实验结果回传不在本文档内。

基本约定：

- 1. 跨系统唯一标识使用 dispatch_id。每次发送都会生成新的 dispatch_id；后续若需对账、查状态或对结果，均以此编号为准。
- 2. order_no 是业务侧订单编号，便于人员识别；业务上允许重复，不能单独作为系统唯一键。
- 3. 通信为同步 HTTP：镁伽系统在同一次调用中返回受理结果。Vita 仅当响应 code=0 时视为本次下发成功。

项	内容
接口名称	接收流式工单
请求方式	POST
请求路径	由镁伽系统提供（待确认）
Content-Type	application/json; charset=UTF-8
调用方	Vita
提供方	镁伽系统

二、请求体结构说明

请求 Body 为单个 JSON 对象。顶层包含下发元数据、PC 信息表、样本板列表、细胞板列表。样本板通过 cell_keys 引用细胞板上的具体列；样本孔通过可选的 pc_id 引用 PC 信息表中的条目。

下表「必填」表示：该字段在 Vita 校验通过并实际下发时必须具备有效值。「示例」仅作取值示意，联调以真实业务数据为准。

2.1 根对象

根对象描述一次下发的基本信息，以及三块核心业务内容：pc_infos、sample_plates、cell_plates。

字段	类型	必填	说明	示例
dispatch_id	string	是	本次发送的唯一编号。每次点击发送都会生成新值，用于后续匹配	DSP260721482913
order_no	string	是	业务订单号，供展示与人工识别；允许业务侧重复	CN2605030000526
order_name	string	否	订单名称；无则传空字符串 ""	CN2605030000526-GPC3
data_type	string	是	样品来源类型。取值：TITER（效价）、PLAS（质粒）、PCR	TITER
priority	string	是	设备侧优先级。取值：high / normal / low	normal
pc_infos	object[]	是	PC 信息表。没有对照信息时传空数组 []	见 2.2
sample_plates	object[]	是	样本板列表。至少包含 1 块样本板	见 2.3
cell_plates	object[]	是	细胞板列表。至少包含 1 块细胞板，且整单至少有一列已填写细胞名称	见 2.5

2.2 根对象.pc_infos[]（PC 信息表）

用于存放阳性对照、同型对照、标签对照等辅助信息。数组本身可以为空。若数组中存在条目，则该条目的 pc_id、pc_type 必须填写。

样本孔是否引用本表，由 wells[].pc_id 决定，并非强制。孔属于何种类型，以 wells[].content_type 为准。

字段	类型	必填	说明	示例
pc_id	string	是	本条 PC 记录的唯一 ID，供孔位可选引用	pc_1001
pc_type	string	是	PC 记录类型。取值：SERUM / ISO / TAG	SERUM
pc_name	string	否	名称	TNFR2-BI1910-hlgG1
catalog_batch	string	否	货号或批次	P220826S01
source	string	否	来源	
concentration	string	否	浓度	

2.3 根对象.sample_plates[]（样本板）

每块样本板描述一块 96 孔板的条码、项目与靶点信息、选用的检测细胞列，以及全部孔位内容。同一请求内样本板条码不可重复，也不得与细胞板条码重复。

字段	类型	必填	说明	示例
barcode	string	是	样本板条码	CN2605030000526-20R168-RMBW-1
project_no	string	是	项目号	CN2605030000526
target	string	是	靶点	GPC3
secondary_antibody	string	否	二抗。Vita 侧缺省为「人」	人
cell_keys	string[]	是	本样本板选用的细胞列列表，至少 1 项。每项格式为「细胞板条码 列号」，例如 GPC3-H-1 1	["GPC3-H-1 1"]
wells	object[]	是	孔位列表。必须恰好 96 条，well_no 覆盖 A01-H12 且不重复	见 2.4

2.4 根对象.sample_plates[].wells[]（样本板孔位）

每个孔必须声明孔号与孔类型。SAMPLE 孔通常填写 sample_code; pc_id 为可选字段，用于引用 pc_infos 中的某条记录，不引用时传 null。

再次强调：孔类型只看 content_type。是否填写 pc_id，以及 pc_infos 中对应条目的 pc_type 是什么，都不改变该孔的 content_type。

字段	类型	必填	说明	示例
well_no	string	是	孔号，取值范围 A01-H12	A01
content_type	string	是	孔类型。取值：SAMPLE / PC / NC / ISO / TAG / BLANK	SAMPLE
sample_code	string	否	样本编号。一般用于 SAMPLE 孔；其他类型可为空字符串	01AAA23G
pc_id	string / null	否	可选引用 pc_infos[].pc_id；不引用时传 null	pc_1001

2.5 根对象.cell_plates[] (细胞板)

每块细胞板固定 12 列 (column_no 为 1-12)。样本板通过 cell_keys 引用「某细胞板条码 + 列号」组合，表示该样本板需要检测哪些细胞列。

字段	类型	必填	说明	示例
barcode	string	是	细胞板条码。同一请求内唯一，且不得与样本板条码重复	GPC3-H-1
columns	object[]	是	列信息数组，必须恰好 12 条，column_no 覆盖 1-12	见 2.6

2.6 根对象.cell_plates[].columns[] (细胞列)

未启用的列将 cell_name 置为空字符串。整单至少需要一列填写 cell_name；当 cell_name 非空时，cell_type 必须填写。batch、generation 等为细胞属性，仅在细胞列上维护。

字段	类型	必填	说明	示例
column_no	number	是	列号，取值 1-12	1
cell_name	string	条件	细胞名称。空字符串表示未启用；整单至少一列非空	CHO-S-hGPC3 (T) - V5
cell_type	string	条件	细胞类型。当 cell_name 非空时必须填	正常
species	string	否	种属	人
batch	string	否	批次	C260107B01
generation	string	否	代数	P14
cell_count	string	否	细胞数	
catalog_no	string	否	货号	
source	string	否	来源	

三、响应说明

镁伽系统应在同一次 HTTP 请求中同步返回 JSON。Vita 根据 code 判断本次下发是否受理成功：仅 code=0 视为成功；非 0 视为失败，本次 dispatch_id 不进入后续执行流程。

3.1 响应字段

字段	类型	必填	说明	示例
code	number	是	业务结果码。0 表示成功，非 0 表示失败	0
message	string	是	结果说明。失败时应给出可读原因，便于排查	success
data	object / null	是	成功时返回受理信息对象；失败时必须为 null	见 3.2

3.2 data (仅当 code=0)

字段	类型	必填	说明	示例
dispatch_id	string	是	回显请求中的 dispatch_id, 确认受理的是哪一次发送	DSP260721482913
accepted	boolean	是	是否已受理。成功时固定为 true	true

3.3 code 取值说明

code	含义	说明
0	成功	已受理本次下发
400	参数错误	缺字段、格式不合法、板/孔结构不完整、条码冲突等
409	重复接收	同一 dispatch_id 已成功接收过
500	服务端异常	镁伽系统内部错误

3.4 成功响应示例

```
{
  "code": 0,
  "message": "success",
  "data": {
    "dispatch_id": "DSP260721482913",
    "accepted": true
  }
}
```

3.5 失败响应示例

```
{
  "code": 400,
  "message": "sample_plates[0].wells 必须包含 A01-H12 共 96 个孔位",
  "data": null
}

{
  "code": 409,
  "message": "dispatch_id 已存在: DSP260721482913",
  "data": null
}
```

四、完整请求示例

以下给出一次完整下发 Body，便于联调对照。示例中的订单号、条码、样本编码、细胞与 PC 名称等为业务侧常见写法示意，正式数据以实际工单为准。

示例内容包括：1 条 PC 信息、1 块样本板（含完整 96 孔）、1 块细胞板（12 列，其中第 1–4 列已启用）。A12 为 PC 孔，并可选地引用了 pc_infos 中的 pc_1001。

```
{
  "dispatch_id": "DSP260721482913",
  "order_no": "CN2605030000526",
  "order_name": "CN2605030000526-GPC3",
  "data_type": "TITER",
  "priority": "normal",
  "pc_infos": [
    {
      "pc_id": "pc_1001",
      "pc_type": "SERUM",
      "pc_name": "TNFR2-BI1910-hIgG1",
      "catalog_batch": "P220826S01",
      "source": "",
      "concentration": ""
    }
  ],
  "sample_plates": [
    {
      "barcode": "CN2605030000526-20R168-RMBW-1",
      "project_no": "CN2605030000526",
      "target": "GPC3",
      "secondary_antibody": "人",
      "cell_keys": [
        "GPC3-H-1|1",
        "GPC3-H-1|2",
        "GPC3-H-1|3",
        "GPC3-H-1|4"
      ],
      "wells": [
        {
          "well_no": "A01",
          "content_type": "SAMPLE",
          "sample_code": "01AAA23G",
          "pc_id": null
        },
        {
          "well_no": "A02",
          "content_type": "SAMPLE",
          "sample_code": "01AAAMUT",
          "pc_id": null
        },
        {
          "well_no": "A03",
          "content_type": "SAMPLE",
          "sample_code": "01AAAMV0",
          "pc_id": null
        }
      ]
    }
  ]
}
```

```
"well_no": "A04",
"content_type": "SAMPLE",
"sample_code": "01AAAMV1",
"pc_id": null
},
{
  "well_no": "A05",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMV2",
  "pc_id": null
},
{
  "well_no": "A06",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMV3",
  "pc_id": null
},
{
  "well_no": "A07",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMV4",
  "pc_id": null
},
{
  "well_no": "A08",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMV5",
  "pc_id": null
},
{
  "well_no": "A09",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMV6",
  "pc_id": null
},
{
  "well_no": "A10",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMV7",
  "pc_id": null
},
{
  "well_no": "A11",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMV8",
  "pc_id": null
},
{
  "well_no": "A12",
  "content_type": "PC",
  "sample_code": "",
  "pc_id": "pc_1001"
},
{
  "well_no": "B01",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMXR",
  "pc_id": null
},
{
  "well_no": "B02",
  "content_type": "SAMPLE",
  "sample_code": "01AAAMXT",
```



```
    "well_no": "C01",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C02",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C03",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C04",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C05",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C06",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C07",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C08",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C09",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C10",
    "content_type": "BLANK",
    "sample_code": "",
    "pc_id": null
  },
  {
    "well_no": "C11",
    "content_type": "BLANK",
    "sample_code": "",
```

```
"pc_id": null
},
{
  "well_no": "C12",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D01",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D02",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D03",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D04",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D05",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D06",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D07",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D08",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D09",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{

```

```
"well_no": "D10",
"content_type": "BLANK",
"sample_code": "",
"pc_id": null
},
{
  "well_no": "D11",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "D12",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E01",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E02",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E03",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E04",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E05",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E06",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E07",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "E08",
  "content_type": "BLANK",
  "sample_code": "",
```



```
"well_no": "F07",
"content_type": "BLANK",
"sample_code": "",
"pc_id": null
},
{
  "well_no": "F08",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "F09",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "F10",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "F11",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "F12",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G01",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G02",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G03",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G04",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G05",
  "content_type": "BLANK",
  "sample_code": "",
```

```
"pc_id": null
},
{
  "well_no": "G06",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G07",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G08",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G09",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G10",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G11",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "G12",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "H01",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "H02",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
  "well_no": "H03",
  "content_type": "BLANK",
  "sample_code": "",
  "pc_id": null
},
{
```

```

        "well_no": "H04",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H05",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H06",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H07",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H08",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H09",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H10",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H11",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    },
    {
        "well_no": "H12",
        "content_type": "BLANK",
        "sample_code": "",
        "pc_id": null
    }
    ]
}
],
"cell_plates": [
    {
        "barcode": "GPC3-H-1",
        "columns": [
            {
                "column_no": 1,
                "cell_name": "CH0-S-hGPC3 (T) -V5",

```

```
"cell_type": "正常",
"species": "人",
"batch": "C260107B01",
"generation": "P14",
"cell_count": "",
"catalog_no": "",
"source": ""
},
{
  "column_no": 2,
  "cell_name": "CH0-S-huGPC3-V5-PDGFR",
  "cell_type": "正常",
  "species": "人",
  "batch": "C260410B01",
  "generation": "P14",
  "cell_count": "",
  "catalog_no": "",
  "source": ""
},
{
  "column_no": 3,
  "cell_name": "CH0-S-mGPC3-V5",
  "cell_type": "正常",
  "species": "人",
  "batch": "C260414B01",
  "generation": "P14",
  "cell_count": "",
  "catalog_no": "",
  "source": ""
},
{
  "column_no": 4,
  "cell_name": "CH0-S-dogGPC3-V5",
  "cell_type": "正常",
  "species": "人",
  "batch": "C260414B01",
  "generation": "P14",
  "cell_count": "",
  "catalog_no": "",
  "source": ""
},
{
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  "source": ""
},
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},
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  "source": ""
},
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  "species": "",
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  "catalog_no": "",
  "source": ""
},
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  "cell_name": "",
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  "species": "",
  "batch": "",
  "generation": "",
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  "catalog_no": "",
  "source": ""
},
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  "cell_name": "",
  "cell_type": "",
  "species": "",
  "batch": "",
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  "catalog_no": "",
  "source": ""
},
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  "cell_name": "",
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  "cell_count": "",
  "catalog_no": "",
  "source": ""
},
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  "column_no": 12,
  "cell_name": "",
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  "species": "",
  "batch": "",
  "generation": "",
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    "cell_count": "",  
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]  
}  
]
```